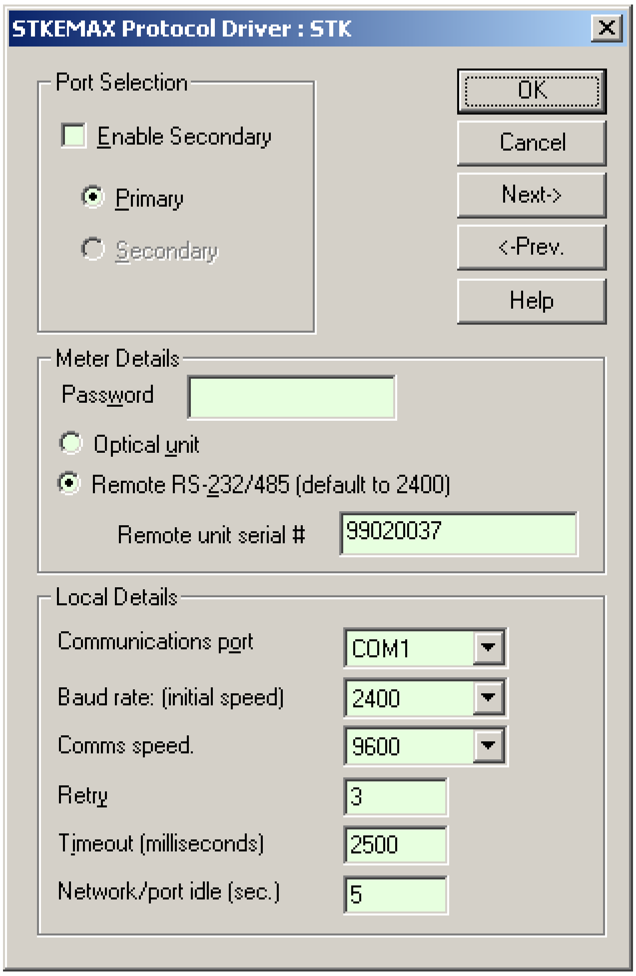
STKEMAX : Strike Technologies EnerMAX

Overview

Adroit supports communication with the Strike Technologies Ener-max power meter.



Meter Details

Password – Password required for any meter controls.

Optical unit – Select this option to use the optical interface unit.

Remote RS-232/485 – Select to use directly cabled option

Serial #– Serial number to identify/address the unit.

Local Details

Port - The available serial ports on your machine

Baud rate – The default or connecting speed for the meter (300 for optical and 2400 if remote).

Comms Speed – The speed the driver must change to after connecting to the meter, the meter returns to the default speed after every transaction.

Retry - The number of times that Adroit will try to gain communication with a meter before failing a transaction.

Timeout - The time that Adroit will wait for a valid response from a FED before failing a transaction (Read or Write).

Network/port Idle – time in seconds to wait between transactions, (values 1-15). This time applies to all Reads and Writes out to the meter.

NOTES: Settings are 1 start, 7 data, even parity and one stop bit. When using Optical unit, the initial baud-rate is always 300, on remote the rate can be changed, but should use 2400.

Wiring

Cable for direct connection to power meter.

PLC 25 pin		Power meter
2	———	RX
3	———	TX
7	———	GND

Supported addresses

Addressable items in the Enermax power meter.

Device	Min Range	Max Range	PLC Data Type	BOOLEAN	INTEGER	REAL
Billing & status values	0	255	Register		BS 0	BS 0
Initiate billing read.	0	0	none	BT 0		
Set clock	0	0	Packed time		ST 0	
Reset billing data	0	0	Result=1 if succesfull else=0 if not	RB 0		
Read profile			UNSUPPORTED			
Clear profile			UNSUPPORTED			
Hex Dump			UNSUPPORTED			
Clear meter registers	0	0	Result=1 if succesfull else=0 if not	CR 0		
Read profile reversed			UNSUPPORTED			
Clear QOS data			UNSUPPORTED			

Reading Billing

Reading of Billing and status “BS” is achieved by toggling a digital scanned to “BT” to the on state and then off (use a pulsed digital), this instructs the driver to update it’s billing database on the next Adroit scan of any of the the Billing & Status “BS” tags. A Expression can thus be used to trigger the retrieval of data at specific times of the day.

If you do not scan “BT” (must be output enabled), polling will operate as normal. I.e. on every scan, the driver will directly fetch fresh BS data.

The “BT” or Billing Trigger is more meaningfull, if your scan-time is short, example 4 seconds (4000ms), since the driver will only poll Billing data on the next poll anyway. A usefull feature of the “BT” control tag, is that you can attach a button to the digital and create a “Refresh now” feature in your user interface or mimic. Remember that if you do not scan a digital to “BT”, the scanning interval works as normal for a polled driver.

Meter Register/data types

The following objects are recognized and parsed:

Date= HH:MM DD/MM/YY

Date2 = HH:MM DDMMYY

Time = HH:MM

Float = number with optional fractional portion following a period mark

Register Listing

Register designations are not shown.

Register	Type :	Register	Type :
0	Time & Date – Running	129	Reverse Wh – Total – Stack 4
1	Forward W/h – Rate 1 – Running	130	Lagging Varh – Rate 1 – Stack 4
2	Forward W/h – Rate 2 – Running	131	Lagging Varh – Rate 2 – Stack 4
3	Forward W/h – Rate 3 – Running	132	Lagging Varh – Rate 3 – Stack 4
4	Forward W/h – Total – Running	133	Lagging Varh – Total – Stack 4
5	Reverse W/h – Total – Running	134	Leading Varh – Total – Stack 4
6	Lagging Varh – Rate 1 – Running	135	Excess Varh – Program – Stack 4
7	Lagging Varh – Rate 2 – Running	136	MD1 – Rate 1 – Stack 4
8	Lagging Varh – Rate 3 – Running	137	MD2 – Rate 2 – Stack 4
9	Lagging Varh – Total – Running	138	MD3 – Rate 3 – Stack 4
10	Leading Varh – Total – Running	139	MD4 – Program – Stack 4
11	Excess Varh – Program – Running	140	Cumilitive MD1 – Rate 1 – Stack 4
12	MD1 – Rate 1 – Running	141	Cumilitive MD2 – Rate 2 – Stack 4
13	MD2 – Rate 2 – Running	142	Cumilitive MD3 – Rate 3 – Stack 4
14	MD3 – Rate 3 – Running	143	Cumilitive MD4 – Program – Stack 4
15	MD4 – Program – Running	144	Excess Varh – Rate 1 – Stack 4
16	Cumilitive MD1 – Rate 1 – Running	145	Excess Varh – Rate 2 – Stack 4
17	Cumilitive MD2 – Rate 2 – Running	146	Excess Varh – Rate 3 – Stack 4
18	Cumilitive MD3 – Rate 3 – Running	147	Vah – Total – Stack 4
19	Cumilitive MD4 – Program – Running	148	MD5 – Program – Stack 4
20	Excess Varh – Rate 1 – Running	149	Time & Date – MD5 – Stack 4
21	Excess Varh – Rate 2 – Running	150	Time & Date – MD4 – Stack 4
22	Excess Varh – Rate 3 – Running	151	Cumulitive MD5 – Program – Stack 4
23	Vah – Total – Running	152	Future Use – Stack 4
24	MD5 – Program – Running	153	Future Use – Stack 4
25	Time & Date – MD5 – Running	154	Future Use – Stack 4
26	Time & Date – MD4 – Running	155	Last Demand MD4
27	Cumulitive MD5 – Program – Running	156	Instantaneous Voltage (L1)
28	Future Use – Running	157	Instantaneous Voltage (L2)
29	Future Use – Running	158	Instantaneous Voltage (L3)
30	Future Use – Running	159	Instantaneous Current (L1)
31	Time & Date – Stack 1	160	Instantaneous Current (L2)
32	Forward Wh – Rate 1 – Stack 1	161	Instantaneous Current (L3)
33	Forward Wh – Rate 2 – Stack 1	162	Instantaneous Real Power
34	Forward Wh – Rate 3 – Stack 1	163	Instantaneous Reactive Power
35	Forward Wh – Total – Stack 1	164	Instantaneous Apparent Power
36	Reverse Wh – Total – Stack 1	165	Instantaneous Power Factor
37	Lagging Varh – Rate 1 – Stack 1	166	Present Thermal Demand – MD4
38	Lagging Varh – Rate 2 – Stack 1	167	Present Thermal Demand – MD5
39	Lagging Varh – Rate 3 – Stack 1	168	Last Demand – MD5
40	Lagging Varh – Total – Stack 1	169	Elepsed Time in block Interval

41	Leading Varh – Total – Stack 1	170	Meter Serial Number
42	Excess Varh – Program – Stack 1	171	Meter Software version and Date
43	MD1 – Rate 1 – Stack 1	172	Meter External VT Ratio
44	MD2 – Rate 2 – Stack 1	173	Meter External CT Ratio
45	MD3 – Rate 3 – Stack 1	174	Number of Resets
46	MD4 – Program – Stack 1	175	Number of Reprograms
47	Cumilitive MD1 – Rate 1 – Stack 1	176	Time & Date of Last Reprogram
48	Cumilitive MD2 – Rate 2 – Stack 1	177	Number of Power Downs
49	Cumilitive MD3 – Rate 3 – Stack 1	178	Time & Date of Last Power Down
50	Cumilitive MD4 – Program – Stack 1	179	Time & Date of Last Power Up
51	Excess Varh – Rate 1 – Stack 1	180	Time & Date of 2nd Last Power Down
52	Excess Varh – Rate 2 – Stack 1	181	Time & Date of 2nd Last Power Up
53	Excess Varh – Rate 3 – Stack 1	182	Time & Date of 3rd Last Power Down
54	Vah – Total – Stack 1	183	Time & Date of 3rd Last Power Up
55	MD5 – Program – Stack 1	184	Time & Date of 4th Last Power Down
56	Time & Date – MD5 – Stack 1	185	Time & Date of 4th Last Power Up
57	Time & Date – MD4 – Stack 1	186	Relay 1 Function
58	Cumulitive MD5 – Program – Stack 1	187	Relay 2 Function
59	Future Use – Stack 1	188	Relay 3 Function and Pulse Value
60	Future Use – Stack 1	189	Relay 4 Function and Pulse Value
61	Future Use – Stack 1	190	VT Ratio Adjust
62	Time & Date – Stack 2	191	CT Ratio Adjust
63	Forward Wh – Rate 1 – Stack 2	192	Enhanced Memory Size
64	Forward Wh – Rate 2 – Stack 2	193	Active Tariff Name
65	Forward Wh – Rate 3 – Stack 2	194	Active rate, day, season and holiday
66	Forward Wh – Total – Stack 2	195	Memory Bank and Activation Day
67	Reverse Wh – Total – Stack 2	196	Recording Channel Status
68	Lagging Varh – Rate 1 – Stack 2	197	Block Interval ,demand & sync mode
69	Lagging Varh – Rate 2 – Stack 2	198	Battery Age
70	Lagging Varh – Rate 3 – Stack 2	199	Diagnostic Word
71	Lagging Varh – Total – Stack 2	200	Reset Mode and Status
72	Leading Varh – Total – Stack 2	201	Instantaneous W (L1) Unscaled
73	Excess Varh – Program – Stack 2	202	Instantaneous W (L2) Unscaled
74	MD1 – Rate 1 – Stack 2	203	Instantaneous W (L3) Unscaled
75	MD2 – Rate 2 – Stack 2	204	Instantaneous Var (L1) Unscaled
76	MD3 – Rate 3 – Stack 2	205	Instantaneous Var (L1) Unscaled
77	MD4 – Program – Stack 2	206	Instantaneous Var (L1) Unscaled (last know register in v3.0)
78	Cumilitive MD1 – Rate 1 – Stack 2	207	FLOAT
79	Cumilitive MD2 – Rate 2 – Stack 2	208	FLOAT
80	Cumilitive MD3 – Rate 3 – Stack 2	209	FLOAT
81	Cumilitive MD4 – Program – Stack 2	210	FLOAT
82	Excess Varh – Rate 1 – Stack 2	211	FLOAT

83	Excess Varh – Rate 2 – Stack 2	212	FLOAT
84	Excess Varh – Rate 3 – Stack 2	213	FLOAT
85	Vah – Total – Stack 2	214	FLOAT
86	MD5 – Program – Stack 2	215	FLOAT
87	Time & Date – MD5 – Stack 2	216	FLOAT
88	Time & Date – MD4 – Stack 2	217	FLOAT
89	Cumilitive MD5 – Program – Stack 2	218	FLOAT
90	Future Use – Stack 2	219	FLOAT
91	Future Use – Stack 2	220	FLOAT
92	Future Use – Stack 2	221	FLOAT
93	Time & Date – Stack 3	222	FLOAT
94	Forward Wh – Rate 1 – Stack 3	223	FLOAT
95	Forward Wh – Rate 2 – Stack 3	224	FLOAT
96	Forward Wh – Rate 3 – Stack 3	225	FLOAT
97	Forward Wh – Total – Stack 3	226	FLOAT
98	Reverse Wh – Total – Stack 3	227	FLOAT
99	Lagging Varh – Rate 1 – Stack 3	228	FLOAT
100	Lagging Varh – Rate 2 – Stack 3	229	FLOAT
101	Lagging Varh – Rate 3 – Stack 3	230	FLOAT
102	Lagging Varh – Total – Stack 3	231	FLOAT
103	Leading Varh – Total – Stack 3	232	FLOAT
104	Excess Varh – Program – Stack 3	233	FLOAT
105	MD1 – Rate 1 – Stack 3	234	FLOAT
106	MD2 – Rate 2 – Stack 3	235	FLOAT
107	MD3 – Rate 3 – Stack 3	236	FLOAT
108	MD4 – Program – Stack 3	237	FLOAT
109	Cumilitive MD1 – Rate 1 – Stack 3	238	FLOAT
110	Cumilitive MD2 – Rate 2 – Stack 3	239	FLOAT
111	Cumilitive MD3 – Rate 3 – Stack 3	240	FLOAT
112	Cumilitive MD4 – Program – Stack 3	241	FLOAT
113	Excess Varh – Rate 1 – Stack 3	242	FLOAT
114	Excess Varh – Rate 2 – Stack 3	243	FLOAT
115	Excess Varh – Rate 3 – Stack 3	244	FLOAT
116	Vah – Total – Stack 3	245	FLOAT
117	MD5 – Program – Stack 3	246	FLOAT
118	Time & Date – MD5 – Stack 3	247	FLOAT
119	Time & Date – MD4 – Stack 3	248	FLOAT
120	Cumilitive MD5 – Program – Stack 3	249	FLOAT
121	Future Use – Stack 3	250	FLOAT
122	Future Use – Stack 3	251	FLOAT
123	Future Use – Stack 3	252	FLOAT
124	Time & Date – Stack 4	253	FLOAT
125	Forward Wh – Rate 1 – Stack 4	254	FLOAT

126	Forward Wh – Rate 2 – Stack 4	255	FLOAT
127	Forward Wh – Rate 3 – Stack 4		
128	Forward Wh – Total – Stack 4		

The driver will interpret all registers as they appear in this table, up to and including # 255 if it is present (driver revision #4).

Message protocol

This driver complies with a subset of the IEC 61107 (formerly 1107) specification and has only been tested in communication with Strike Enermax power meters. It does not implement any encryption/decryption features.

Communication follows a 3-step cycle.

INITIATE REQUEST (OPTICAL):

The initiate step is common to all transactions, this is a log-on.

Initiate request (local form)

Start character	Request code	End	Completion charcters
/	?	!	<CR><LF>

Initiate request (remote form)

Start character	Request code	Address	End	Completion charcters
/	?	12345678	!	<CR><LF>

Initiate response:

Start	Suggest rate	Serial # (if remote)	End	Completion characters
/STR	5	02	!	<CR><LF>

READ BILLING DATA REQUEST:

Follows on from initiate step.

Start character	Request code	New Baud Rate	Fn.	Completion charcters
x06	0	3	0	<CR><LF>

Read billing data response:

Start charcter	Data	End	Completion characters	ETX	CRC
x01	??	!	<CR><LF>	03	XX XX

Data is in following format, one register per line

Register #	Start	Data	End	Completion characters
000	'('	Text value	')'	<CR><LF>

logoff is automatic.

SET CLOCK:

Start character	Request code	New Baud Rate	Fn.	Completion charcters
x06	0	3	1	<CR><LF>

Password prompt:

Start charcter	Function	STX	STX	Completion characters	ETX	CRC
x01	P0	x02	x02	<CR><LF>	03	XX XX

Example of datascope follows here

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~TX: Selected 2400 baud./?99020037!
~RX: /STR5990
~RX: 20037V4.1EM1
~TX: 050
~TX: Selected 9600 baud.
~RX: 000(13:37 08/04/02*) - OK [1018265820.000]
~RX: 001(00000.0*kWh) - OK [0.000]
~RX: 002(00000.0*kWh) - OK [0.000]
~RX: 003(00000.0*kWh) - OK [0.000]
~RX: 004(00000.0*kWh) - OK [0.000]
~RX: 005(00000.4*kWh) - OK [0.400]
~RX: 006(00000.0*kvarh) - OK [0.000]
~RX: 007(00000.0*kvarh) - OK [0.000]
~RX: 008(00000.0*kvarh) - OK [0.000]
~RX: 009(00000.0*kvarh) - OK [0.000]
~RX: 010(00000.6*kvarh) - OK [0.600]
~RX: 011(00000.0*kvarh) - OK [0.000]
~RX: 012(00.0004*kVA) - OK [0.000]
~RX: 013(00.0000*kVA) - OK [0.000]
~RX: 014(00.0000*kVA) - OK [0.000]
~RX: 015(00.0004*kVA) - OK [0.000]
~RX: 016(00.0004*kVA) - OK [0.000]
~RX: 017(00.0000*kVA) - OK [0.000]
~RX: 018(00.0000*kVA) - OK [0.000]
~RX: 019(00.0004*kVA) - OK [0.000]
~RX: 020(00000.0*kvarh) - OK [0.000]
~RX: 021(00000.0*kvarh) - OK [0.000]
~RX: 022(00000.0*kvarh) - OK [0.000]
~RX: 023(00000.0*kVAh) - OK [0.000]
~RX: 024(00.0004*kVA) - OK [0.000]
~RX: 025(16:30 04/03/99*) - OK [920557800.000]
~RX: 026(16:30 04/03/99*) - OK [920557800.000]
~RX: 027(00.0004*kVA) - OK [0.000]
~RX: 028(Future Use*) - OK [0.000]
~RX: 029(Future Use*) - OK [0.000]
~RX: 030(Future Use*) - OK [0.000]
~RX: 031(15:44 04/03/99*) - OK [920555040.000]
~RX: 032(00000.0*kWh) - OK [0.000]
~RX: 033(00000.0*kWh) - OK [0.000]
~RX: 034(00000.0*kWh) - OK [0.000]
~RX: 035(00000.0*kWh) - OK [0.000]
~RX: 036(00000.0*kWh) - OK [0.000]
~RX: 037(00000.0*kvarh) - OK [0.000]
~RX: 038(00000.0*kvarh) - OK [0.000]
~RX: 039(00000.0*kvarh) - OK [0.000]
~RX: 040(00000.0*kvarh) - OK [0.000]
~RX: 041(00000.0*kvarh) - OK [0.000]
~RX: 042(00000.0*kvarh) - OK [0.000]
~RX: 043(00.0000*kVA) - OK [0.000]
~RX: 044(00.0000*kVA) - OK [0.000]
~RX: 045(00.0000*kVA) - OK [0.000]
~RX: 046(00.0000*kVA) - OK [0.000]
~RX: 047(00.0000*kVA) - OK [0.000]
~RX: 048(00.0000*kVA) - OK [0.000]
~RX: 049(00.0000*kVA) - OK [0.000]
~RX: 050(00.0000*kVA) - OK [0.000]
~RX: 051(00000.0*kvarh) - OK [0.000]
~RX: 052(00000.0*kvarh) - OK [0.000]
~RX: 053(00000.0*kvarh) - OK [0.000]
~RX: 054(00000.0*kVAh) - OK [0.000]
~RX: 055(00.0000*kVA) - OK [0.000]
~RX: 056(00:00 00/00/00*) - OK [0.000]
~RX: 057(00:00 00/00/00*) - OK [0.000]
~RX: 058(00.0000*kVA) - OK [0.000]
~RX: 059(Future Use*) - OK [0.000]
~RX: 060(Future Use*) - OK [0.000]
~RX: 061(Future Use*) - OK [0.000]
~RX: 062(15:43 03/03/99*) - OK [920468580.000]
~RX: 063(00000.0*kWh) - OK [0.000]
~RX: 064(00000.0*kWh) - OK [0.000]
~RX: 065(00000.0*kWh) - OK [0.000]
~RX: 066(00000.0*kWh) - OK [0.000]
~RX: 067(00000.0*kWh) - OK [0.000]
~RX: 068(00000.0*kvarh) - OK [0.000]
~RX: 069(00000.0*kvarh) - OK [0.000]
~RX: 070(00000.0*kvarh) - OK [0.000]
~RX: 071(00000.0*kvarh) - OK [0.000]
~RX: 072(00000.0*kvarh) - OK [0.000]
~RX: 073(00000.0*kvarh) - OK [0.000]
~RX: 074(00.0000*kVA) - OK [0.000]
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~RX: 075(00.0000*kVA) - OK [0.000]
~RX: 076(00.0000*kVA) - OK [0.000]
~RX: 077(00.0000*kVA) - OK [0.000]
~RX: 078(00.0000*kVA) - OK [0.000]
~RX: 079(00.0000*kVA) - OK [0.000]
~RX: 080(00.0000*kVA) - OK [0.000]
~RX: 081(00.0000*kVA) - OK [0.000]
~RX: 082(00000.0*kvarh) - OK [0.000]
~RX: 083(00000.0*kvarh) - OK [0.000]
~RX: 084(00000.0*kvarh) - OK [0.000]
~RX: 085(00000.0*kVAh) - OK [0.000]
~RX: 086(00.0000*kVA) - OK [0.000]
~RX: 087(00:00 00/00/00*) - OK [0.000]
~RX: 088(00:00 00/00/00*) - OK [0.000]
~RX: 089(00.0000*kVA) - OK [0.000]
~RX: 090(Future Use*) - OK [0.000]
~RX: 091(Future Use*) - OK [0.000]
~RX: 092(Future Use*) - OK [0.000]
~RX: 093(15:18 04/03/99*) - OK [920553480.000]
~RX: 094(00000.0*kWh) - OK [0.000]
~RX: 095(00000.0*kWh) - OK [0.000]
~RX: 096(00000.0*kWh) - OK [0.000]
~RX: 097(00000.0*kWh) - OK [0.000]
~RX: 098(00000.0*kWh) - OK [0.000]
~RX: 099(00000.0*kvarh) - OK [0.000]
~RX: 100(00000.0*kvarh) - OK [0.000]
~RX: 101(00000.0*kvarh) - OK [0.000]
~RX: 102(00000.0*kvarh) - OK [0.000]
~RX: 103(00000.0*kvarh) - OK [0.000]
~RX: 104(00000.0*kvarh) - OK [0.000]
~RX: 105(00.0000*kVA) - OK [0.000]
~RX: 106(00.0000*kVA) - OK [0.000]
~RX: 107(00.0000*kVA) - OK [0.000]
~RX: 108(00.0000*kVA) - OK [0.000]
~RX: 109(00.0000*kVA) - OK [0.000]
~RX: 110(00.0000*kVA) - OK [0.000]
~RX: 111(00.0000*kVA) - OK [0.000]
~RX: 112(00.0000*kVA) - OK [0.000]
~RX: 113(00000.0*kvarh) - OK [0.000]
~RX: 114(00000.0*kvarh) - OK [0.000]
~RX: 115(00000.0*kvarh) - OK [0.000]
~RX: 116(00000.0*kVAh) - OK [0.000]
~RX: 117(00.0000*kVA) - OK [0.000]
~RX: 118(00:00 00/00/00*) - OK [0.000]
~RX: 119(00:00 00/00/00*) - OK [0.000]
~RX: 120(00.0000*kVA) - OK [0.000]
~RX: 121(Future Use*) - OK [0.000]
~RX: 122(Future Use*) - OK [0.000]
~RX: 123(Future Use*) - OK [0.000]
~RX: 124(00:00 00/00/00*) - OK [0.000]
~RX: 125(00000.0*kWh) - OK [0.000]
~RX: 126(00000.0*kWh) - OK [0.000]
~RX: 127(00000.0*kWh) - OK [0.000]
~RX: 128(00000.0*kWh) - OK [0.000]
~RX: 129(00000.0*kWh) - OK [0.000]
~RX: 130(00000.0*kvarh) - OK [0.000]
~RX: 131(00000.0*kvarh) - OK [0.000]
~RX: 132(00000.0*kvarh) - OK [0.000]
~RX: 133(00000.0*kvarh) - OK [0.000]
~RX: 134(00000.0*kvarh) - OK [0.000]
~RX: 135(00000.0*kvarh) - OK [0.000]
~RX: 136(00.0000*kVA) - OK [0.000]
~RX: 137(00.0000*kVA) - OK [0.000]
~RX: 138(00.0000*kVA) - OK [0.000]
~RX: 139(00.0000*kVA) - OK [0.000]
~RX: 140(00.0000*kVA) - OK [0.000]
~RX: 141(00.0000*kVA) - OK [0.000]
~RX: 142(00.0000*kVA) - OK [0.000]
~RX: 143(00.0000*kVA) - OK [0.000]
~RX: 144(00000.0*kvarh) - OK [0.000]
~RX: 145(00000.0*kvarh) - OK [0.000]
~RX: 146(00000.0*kvarh) - OK [0.000]
~RX: 147(00000.0*kVAh) - OK [0.000]
~RX: 148(00.0000*kVA) - OK [0.000]
~RX: 149(00:00 00/00/00*) - OK [0.000]
~RX: 150(00:00 00/00/00*) - OK [0.000]
~RX: 151(00.0000*kVA) - OK [0.000]
~RX: 152(Future Use*) - OK [0.000]
~RX: 153(Future Use*) - OK [0.000]
~RX: 154(Future Use*) - OK [0.000]
~RX: 155(00.0004 kVA  *) - OK [0.000]
~RX: 156(L1  222.9  V *) - OK [222.900]
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~RX: 157(L2  0.000  V *) - OK [0.000]
~RX: 158(L3  0.000  V *) - OK [0.000]
~RX: 159(L1  0.000  A *) - OK [0.000]
~RX: 160(L2  0.000  A *) - OK [0.000]
~RX: 161(L3  0.000  A *) - OK [0.000]
~RX: 162( 0.000  W   *) - OK [0.000]
~RX: 163( 0.000  var *) - OK [0.000]
~RX: 164( 0.000  VA  *) - OK [0.000]
~RX: 165(PF  0.999    *) - OK [0.999]
~RX: 166( 0.000  VA  *) - OK [0.000]
~RX: 167( 0.000  VA  *) - OK [0.000]
~RX: 168(00.0004 kVA  *) - OK [0.000]
~RX: 169(ELAPSED 05:20*) - OK [0.000]
~RX: 170(S/N 99020037*) - OK [99020037.000]
~RX: 171(S/W V4.1 1198*) - OK [4.100]
~RX: 172(VT    400/400*) - OK [400.000]
~RX: 173(CT       5/5  *) - OK [5.000]
~RX: 174(RESETS 03    *) - OK [3.000]
~RX: 175(RE-PROGRAM 01*) - OK [0.000]
~RX: 176(09h51 130199P*) - OK [916213860.000]
~RX: 177(POWERDOWNS 07*) - OK [7.000]
~RX: 178(08h28 220301D*) - OK [985242480.000]
~RX: 179(13h31 080402U*) - OK [1018265460.000]
~RX: 180(11h30 120799D*) - OK [931771800.000]
~RX: 181(13h07 200301U*) - OK [985086420.000]
~RX: 182(13h50 280699D*) - OK [930570600.000]
~RX: 183(14h14 280699U*) - OK [930572040.000]
~RX: 184(10h46 040599D*) - OK [925807560.000]
~RX: 185(13h40 280699U*) - OK [930570000.000]
~RX: 186(1 C         U*) - OK [0.000]
~RX: 187(2 C         U*) - OK [0.000]
~RX: 188(3 0.0010k    *) - OK [0.001]
~RX: 189(4 0.0010k    *) - OK [0.001]
~RX: 190(VT ADJ 0.00%*) - OK [0.000]
~RX: 191(CT ADJ 0.00%*) - OK [0.000]
~RX: 192(MEM  5 x 32k *) - OK [5.000]
~RX: 193(FACTORY SETUP*) - OK [0.000]
~RX: 194(R1;S1;D1;    *) - OK [1.000]
~RX: 195(BANK 2;      *) - OK [2.000]
~RX: 196(RecCh 1;0;1;0*) - OK [1.000]
~RX: 197(BLOCK 30;VA;I*) - OK [30.000]
~RX: 198(BAT AGE 03.2y*) - OK [3.200]
~RX: 199(ERROR FREE   *) - OK [0.000]
~RX: 200(RSET MAN PO; *) - OK [0.000]
~RX: 201(L1  0.000  W *) - OK [0.000]
~RX: 202(L2  0.000  W *) - OK [0.000]
~RX: 203(L3  0.000  W *) - OK [0.000]
~RX: 204(L1  0.000 var*) - OK [0.000]
~RX: 205(L2  0.000 var*) - OK [0.000]
~RX: 206(L3  0.000 var*) - OK [0.000]
~RX: !%Transaction completed OK (25106 ms)
```

Troubleshooting

If you see “Billing and status read not flagged, skipping read.” This simply means that Adroit polled for a value, but the “BT 0” digital that was scanned has not yet gone high. This digital needs to be pulsed as well.

Registry keys

When using a Ethernet port-extender, it may be desirable to turn off the communications port restart logic in the driver. When enabled, the driver closes the hardware port, and re-opens it after a delay equal to the timeout setting. This causes problems in some port emulations. Change the setting

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\STKEMAX\STK1\Primary
DWORD ClosePortOnFailure = 1 to zero (0) to prevent the reset logic from running.

Revision

Date	Changes
8/4/02	Added datascope sample, and Troubleshooting section, corrected the

	Scanning address for the trigger to BT 0.
14/7/04	New Dialog screen-shot with help button added in rev 6 of the driver.
21/6/05	Registry key in rev 7 added to allow port resetting to be disabled.
12/10/05	Mention IEC 61107 protocol relationship in protocol section.